Local Competition and Pizza Prices

Group 9 – Uber Eats listings around Houston

1 Economic question

Do restaurants with more nearby pizza competitors list lower pizza prices, after accounting for location, identified chains and menu characteristics?

More nearby sellers give customers alternatives, which can constrain prices. We therefore expected a negative association. In busy areas, strong demand may attract more sellers and sustain higher prices at the same time. Our regressions describe how prices differ across listings with different numbers of neighbours.

2 Data and construction of the outcome

Two linked files

The Uber Eats dataset [1] contains 63,469 US restaurant records with names, categories, addresses, ratings and coordinates. The Houston-area menu extract contains 54,778 rows with item names, categories, descriptions and listed prices. Restaurant ID links the files. A listing is a platform record, a site groups listings with the same rounded coordinates, and an offer is one menu item. The completeness of the regional extract has not been checked against the national menu file.

Which offers count as pizzas

We start from menu categories that mention pizza and require a positive price in US dollars. In mixed sections such as Pizza and Pasta, the item name must identify a pizza or a pizza-style pie. Pot pies and other non-pizza pies are excluded. Slices, bundles, kits, sides and topping supplements are excluded. Whole pizzas such as One Topping Pizza remain.

We look for stated inch sizes in the title, category and description. Multiple conflicting sizes or ranges leave the diameter unknown. Known diameters must lie between 8 and 18 inches. We exclude explicitly square or rectangular formats, including stated two-dimensional sizes. Identical offers within a listing, based on cleaned name, price and diameter, count once.

Mean price by listing

We give equal weight to each retained offer. If listing i has n_i retained offers with prices p_{ij}, its price is

$$P_i = \frac{1}{n_i} \sum_{j=1}^{n_i} p_{ij}.$$

The regressions use $\log P_i$. The mean share of retained offers with a known diameter across listings is 24.1%. Missing diameters limit comparisons across menus even after accounting for observed size information. The 7,934 retained offers in the wider supplier area yield an analysis sample of 596 listings at 551 sites within 50 km of central Houston. The cleaning audit appears in the Appendix.

3 Competition from latitude and longitude

Let ϕ_i and λ_i denote latitude and longitude in radians. For two listings i, j, their Haversine distance is

$$a = \sin^2\left(\frac{\phi_j - \phi_i}{2}\right) + \cos(\phi_i)\cos(\phi_j)\sin^2\left(\frac{\lambda_j - \lambda_i}{2}\right),$$

$$d_{ij} = 2R \arcsin\sqrt{a},$$

where R is approximately 6,371 km. We count distinct sites with $d_{ij} \leq 2$ km after excluding the listing's own site. Coordinates rounded to four decimal places define sites. The 2 km radius approximates local proximity. Actual delivery zones are unobserved.

The focal sample lies within 50 km of $(29.7604, -95.3698)$. We retain pizza suppliers out to 55 km so the outer edge of the focal sample has coverage for the 5 km robustness check. Listings with unusable prices can still contribute to competition. Supplier eligibility uses pizza categories or pizza-menu evidence. The same supplier universe is used in the initial and corrected versions.

4 Model sequence

For M1–M4 we regress $\log P_i$ on a constant, a count of rival sites and successive sets of controls. All models use the same 596 listings in the corrected sample.

Table 1: Model specifications

Model	Regressors added at each step
M1	Count of other pizza sites within 2 km
M2	East and north coordinates, their squares and their product
M3	Indicator for a chain identified from the listing name or menu
M4	Mean known diameter, indicator for no known diameter, share of known diameters and shares labelled small or large

Coordinates are in units of 10 km. Mean known diameter is centred at 12 inches and set to zero after centring where all diameters are missing. The missing-diameter indicator distinguishes those cases. M5 adds one interaction between competition and the chain indicator to M4.

Table 2: Main regression results

Model	Slope on count	HC3 SE	Adjusted R^2	Test RMSE (USD)
M1	0.00622	0.00243	0.008	4.70
M2	0.00498	0.00262	0.019	4.62
M3	0.00489	0.00264	0.018	4.62
M4	0.00494	0.00233	0.234	3.80

Menu controls give the largest increase in adjusted R^2. For the internal prediction check, we fit on 475 listings and evaluate on 121 held-out listings. Shared listings keep their initial split assignments. Dollar predictions use the mean exponentiated training residual to adjust the log-to-dollar transformation. Nearby listings and related brands can appear in both sets.

5 Main result and coefficient tests

In M4, the competition coefficient is 0.00494 with an HC3 standard error of 0.00233. Exponentiating the coefficient and subtracting one gives a +0.50% comparison in fitted geometric price per additional rival site, holding the included controls fixed. The HC3 95% interval is $[+0.04\%, +0.96\%]$ and $p = 0.0346$. Applied to a reference pizza price of \$15, this percentage corresponds to about \$0.07 per additional site. This dollar figure is an illustration of scale.

Table 3: Coefficient and joint tests

Test	Model	Null hypothesis	Statistic	p
T1	M4	Competition coefficient = 0	$t = 2.118$	0.0346
T2	M4	All 11 control coefficients = 0	$F = 23.928$	< 0.0001
T3	M4	All five menu coefficients = 0	$F = 44.389$	< 0.0001
T4	M5	Chain slope difference = 0	$F = 0.661$	0.4166

T1 and T2

T1 is a two-sided t test of the competition slope. At the 5% threshold, the HC3 test rejects zero and the estimated association is positive. T2 is a joint HC3 Wald F test of the 11 controls. The rejection shows that the controls jointly add explanatory information.

T3 and the menu block

T3 jointly tests the five menu coefficients in M4 with HC3 covariance. Under the null, the menu block adds no explanatory information relative to M3. We reject the joint null. This result is consistent with the increase in adjusted R^2 when the menu variables enter.

M5 and T4

Chains could respond less to local competition if some pricing decisions are made centrally. The degree of centralisation can vary across brands and locations. With C_i the competition count and D_i the chain indicator, M5 retains both main effects and adds $\delta C_i D_i$. The competition slope is β for the other or unclassified group and $\beta + \delta$ for identified chains.

The estimated per-site comparisons are +0.64% for the other or unclassified group and +0.20% for identified chains. T4 tests $\delta = 0$ directly. Its p value is 0.4166, so the data provide insufficient evidence of different slopes. The chain classification identifies observable brands. It gives no direct information on how a specific outlet sets prices.

6 Robustness and limitations

Geographic clustering, alternative radii and the $115 offer

The table keeps the M4 controls. The grid-clustered row uses 60 occupied squares of 10 km by 10 km. It allows error dependence within each square. Nearby observations on opposite sides of a grid boundary can still be related.

Table 4: Robustness checks

M4 variant	Per-site comparison	95% interval	p
2 km, HC3	+0.50%	[+0.04%, +0.96%]	0.0346
2 km, 10 km grid clusters	+0.50%	[-0.05%, +1.04%]	0.0736
1 km, HC3	+0.86%	[-0.03%, +1.75%]	0.0587
3 km, HC3	+0.39%	[+0.03%, +0.76%]	0.0354
5 km, HC3	+0.28%	[+0.07%, +0.49%]	0.0086
2 km, without $115 offer	+0.48%	[+0.02%, +0.94%]	0.0389

The 2 km point estimate stays the same with grid clustering, while the interval includes zero and $p = 0.0736$. The alternative radii all give positive estimates. At 1 km the interval includes zero. A one-site difference has a different meaning at each radius because the count covers a different area.

The source lists the Signature Gluten-Free Cheese Pizza at $115 for F#ck Gluten (Fairdale). Other pizza prices for this listing are much lower. We could not independently confirm the historical $115 entry, so the main estimate uses the recorded value. After excluding just that offer and recalculating price and menu controls, the per-site comparison is +0.48%.

Diagnostics

Breusch–Pagan gives $p = 0.0002$ for constant error variance. The test assumes independent observations. We use HC3 for inference with unequal error variances and show geographic clustering as a separate check. RESET gives $p = 0.0285$ for jointly adding the square and cube of fitted log prices. This is evidence of possible misspecification. It does not identify the missing term or the right model.

Interpretation

The estimates compare listed prices in a cross-section of Uber Eats records. Demand, rents, quality and restaurant location choices may affect both the number of nearby sellers and prices. Site grouping is approximate, menu sizes

are often missing, and coverage reflects the available regional extract.

7 Conclusion

The corrected M4 estimate is small and positive. The evidence for a nonzero association weakens when we allow for within-square dependence. Menu controls improve fit, and the chain interaction remains imprecise. The data leave open how prices would respond to a new competitor entering.

References

[1] Ahmed Shahriar Sakib. *Uber Eats USA Restaurants and Menus*. Kaggle. Source files dated 22 November 2023. https://www.kaggle.com/datasets/ahmedshahriarsakib/uber-eats-usa-restaurants-menus.